

Multiple Choice

1. **Answer:** B

Options: A) a; B) Chemistry; C) 0; D) 1

Note: The correct answer is "Chemistry" because in Python, lists are zero-indexed, meaning the index 1 corresponds to the second element of the list, which is "Chemistry".

2. **Answer:** A

Options: A) $O(1)$; B) $O(n)$; C) $O(n^2)$; D) $O(\log n)$

Note: $O(1)$ is the smallest asymptotic notation because it represents a constant time complexity that does not grow with the size of the input, indicating that the algorithm's performance remains constant regardless of input size.

3. **Answer:** A

Options: A) n^m ; B) $n!/(n - m)!$; C) $n!$; D) $n!/(m!(n - m)!)$

Note: The correct answer is n^m because each of the m elements in set A can be independently mapped to any of the n elements in set B, resulting in n choices for each of the m elements, leading to a total of n^m distinct functions.

4. **Answer:** A

Options: A) ['Hi!', 'Hi!', 'Hi!', 'Hi!']; B) ['Hi!'] * 4; C) Error; D) None of the above.

Note: The expression ['Hi!'] * 4 in Python 3 creates a list that contains four references to the same string 'Hi!', resulting in the output ['Hi!', 'Hi!', 'Hi!', 'Hi!'].

5. **Answer:** D

Options: A) 53; B) 29; C) 50; D) 100

Note: The correct answer is 100 because evaluating the recursive function $f(4)$ involves calculating $f(3)$, $f(2)$, and $f(1)$ based on the defined conditions, leading to the final computation of $f(4) = 4 * (f(3) + 4) = 100$ after performing the necessary recursive calculations.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: In Python, negative indexing allows access to elements from the end of a list, so the expression ['a', 'Chemistry', 0, 1][-3] refers to the second element of the list, which is 'Chemistry'.

7. **Answer:** True

Options: A) True; B) False

Note: This statement is true because the set of all bit strings with more ones than zeros

cannot be recognized by a finite automaton, which is the basis for regular expressions, as this language requires counting and maintaining a balance between the number of ones and zeros.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because finding the longest simple cycle in an undirected graph is an NP-hard problem, meaning it cannot be solved in polynomial time unless $P=NP$.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the function `max(l)` correctly identifies and returns the maximum value from the list `[1, 2, 3, 4]`, which is indeed 4.

10. **Answer:** True

Options: A) True; B) False

Note: Merge sort consistently divides the input data into smaller subarrays and sorts them independently, resulting in a time complexity of $O(n \log n)$ regardless of the initial order of the data, unlike algorithms such as quicksort or bubble sort that can perform significantly worse on certain input arrangements.

Long Form Response

11. **Answer:** `def adjac(ele, sub = []): | def get_coordinates(test_tup): | return (res)`

Note: The provided answer correctly defines a function that utilizes recursion to traverse and extract all adjacent coordinates from a given coordinate tuple, thereby addressing the requirements of the question.

12. **Answer:** `def is_Monotonic(A): | return (all(A[i] <= A[i + 1] for i in range(len(A) - 1)) or`

Note: The provided function accurately checks if the array is monotonic by using a generator expression to verify that all elements are either non-decreasing or non-increasing throughout the array.

End of Answer Key